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ORIGINAL ARTICLE

MCFL: multi-label contrastive focal loss for deep imbalanced pedestrian attribute recognition

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Abstract

Pedestrian Attribute Recognition (PAR) can provide valuable clues for several innovative surveillance applications. It is also a difficult task because inference of the multiple attributes at a far distance is challenging in real complex scenarios. Most existing methods improve the PAR with visual attention mechanisms or body-part detection modules, which increase the complexity of networks and require manual annotations of the human body. Also, uneven data distribution, leading to a decline in recall values, is still underestimated. This paper presents a novel multi-label optimization algorithm to mitigate these issues, named Multi-label Contrastive Focal Loss (MCFL). Specifically, we first propose a multi-label focal loss to emphasize the error-prone and minority attributes with a separated re-weighting scheme. And then, we introduce a multi-label contrastive learning strategy based on the multi-label divergences to help the deep network to distinguish the hard fine-grained attributes. We conduct extensive experiments on seven PAR benchmarks, and results indicate that the proposed MCFL with the native ResNet-50 backbone surpasses the state-of-the-art comparison methods in mean accuracy and recall.

Keywords Pedestrian attribute recognition · Multi-label contrastive loss · Deep convolutional neural network · Multi-label learning · Imbalanced learning

1 Introduction

Pedestrian Attribute Recognition (PAR) can provide detailed soft-biometrics and important semantic information for many applications, *i.e.*, person re-identification (re-ID) and person retrieval [1–4]. Unlike the common visual recognition that is known as a single-label task, PAR is a multi-label recognition task [5], which needs to provide several humanly searchable descriptions for one target people, such as hair-length, gender, hat, or glass, as well as other appearance characteristics [6]. It could help generate high-level semantic information or locate specific targets in intelligent video surveillance [1, 7].

Although tremendous progress has been achieved in image recognition, PAR is quite challenging in unconstrained scenarios because the images collected from real surveillance scenarios vary in illumination, camera angles, or human poses. Furthermore, some subtle attributes (*e.g.*, wearing glass, hat, bags and shoes) have more practical significance for retrieval mission but are not easy to recognize at a far distance, as only a small part of pixels are meaningful for the specific attributes (Fig. 1a).

Many studies on this topic, especially deep constitutional neural networks (CNNs) based approaches, are developed to find more discriminative local features with additional attention mechanisms, *e.g.*, Da-HAR [8] and DIAA [9], or extra body-part location modules, *e.g.*, A-AOG [10] and PGDM [11]. However, the computational costs of deep CNNs could be significantly increased by

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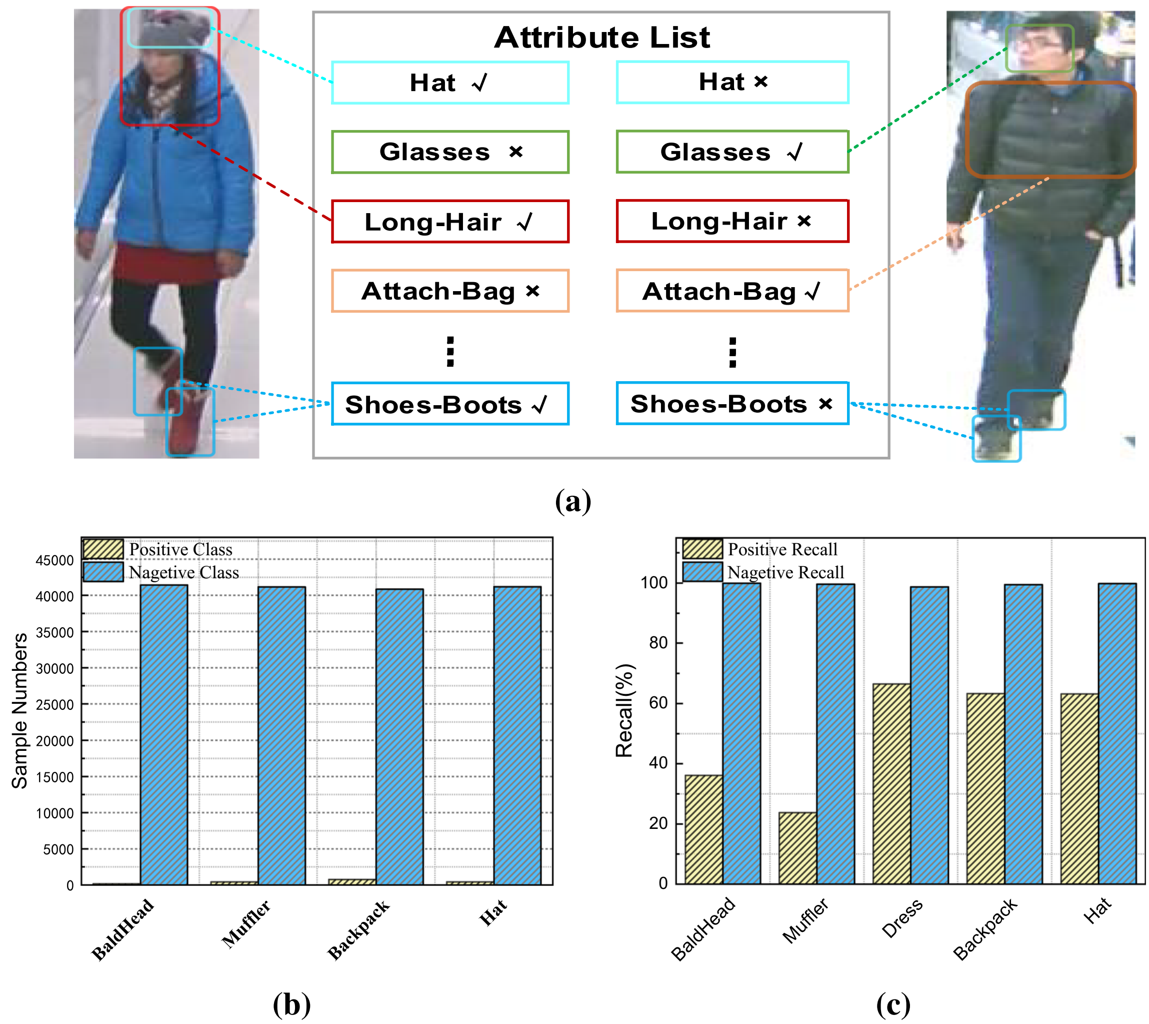


Fig. 1 The example of PAR on RAP dataset. **a** Fine-grained attributes are hard to recognize at far distances. **b** The number of positive attribute samples is far less than that of negative ones. **c** The PAR model is biased towards majority negative classes, resulting in a lower rate of positive recall

using such complex attention or body-part location modules. In addition, annotating large amounts of human body parts is tedious and time-consuming. However, less attention has been paid to enhance PAR with the improved loss function based on the naive CNNs backbones.

Meanwhile, the inherent characteristic of uneven distribution of the PAR data, definitely hindering the performance of attribute retrieval, is often neglected. Fig. 1b shows the data distributions of the positive and negative attributes in PAR dataset [5], respectively. We can infer that the class of interest (positive attributes) often has fewer samples. The uneven nature of the distribution may cause a potential learning bias since the training losses are easily dominated by the majority of negatives samples. It is more difficult for CNNs to distinguish these under-represented attributes, resulting in relatively lower positive recall values. As illustrated in Fig. 1c, the existing model [5] performs much worse on those positive minority classes.

This study presents a novel learning algorithm, named Multi-label Contrastive Focal Loss (MCFL), to address the imbalanced and hard attributes recognition in two folds: a) The error-prone and minority attributes are emphasized through a separated re-weighting focal loss strategy. b) More distinctive features for fine-grained attributes can be extracted by using multi-label contrastive learning without any attributes location modules.

Our main contributions consist in:

(1) We derive an effective optimization method with a more proper training loss function instead of modifying the network architecture of CNNs to improve the PAR.
(2) We develop a multi-label contrastive focal loss, which emphasizes the error-prone and minority-class

samples with large decision margins to distinguish the fine-grained attributes.

(3) We demonstrate the effectiveness of the proposed model on several public PAR datasets. The proposed MCFL based on the native ResNet-50 surpasses the tested comparison approaches in mean accuracy and recall.

A preliminary version of our work first appeared in [12]. We have extended it in the following aspects. (1) Reforming the contrastive loss function to improve the PAR results further. (2) Validating on two additional datasets. (3) Comparison with more recent state-of-the-art PAR approaches. (4) Giving more detailed experimental results and discussion. (5) Conducting extensive experiments to analyze the sensitivity of backbones and hyper-parameters.

The rest part of the paper is organized as follows. The related studies are introduced in Sect. 2. The proposed MCFL method is present in detail in Sect. 3. Experimental results are listed and discussed in Sect. 4, along with the conclusion in Sect. 5.

2 Related work

2.1 Pedestrian attribute recognition

Many deep CNNs-based approaches have been developed to boost PAR recently. These approaches can be categorized as global-based [13, 14], part-based [11, 15, 16], visual attention-based [9, 17–19] and relation-based [20, 20, 21] algorithms. The related studies are introduced as follows.

(a) **Global-based**: DeepMar [5] considers whole images as input and classifies all the attributes simultaneously with a shared backbone CNNs. MTA-Net [22] explores the relations between whole images and attributes using the knowledge of multiple time steps. Global-based models used the features extracted from the whole image ignoring the specific features of each attribute.

(b) **Part-based**: Part-based model focuses on the human body parts to extract the specific local features. PGDM [11] uses a body key-points estimation model to extract the part regions features. A-AOG [10] further parses an input image into different fine-grained components, including each articulation of body parts, and jointly outputs the human detection, pose estimation, and PAR results. However, the Part-based model usually needs auxiliary body-parts information.

(c) **Attention-based**: The attention module can help the CNNs obtaining the specific attributes features on the salient regions. HydraPlus-Net [17] utilizes a multi-directional attention network to obtain multi-level semantic information. Da-HAR [8] develops a coarse-to-fine attention mechanism for localizing attributes and eliminating irrelevant regions. CAS-SAL-FR [23] proposes a cascaded split-and-aggregate (CAS-SAL) model to learn both the individual and common features of attributes, and designs an attribute specific attention mechanism to locate important pixels or regions. However, such attention modules significantly increase the model complexity and computational cost.

(d) **Relation-based**: The Relation-based models utilize the semantic relationship among multi-label to improve the PAR. RCRA [24] integrates two relation modules, *i.e.*, Recurrent convolution (RC) and Recurrent Attention (RA), where RC finds the relationship of inter-group between attributes, and RA explores the intra-group attention locality. JLAC [25] employs Graph Convolutional Network (GCN) to explore the semantic relationships among the attribute-specific nodes. CGCN [26] employs the GCN to construct the multi-layer of attribute relations and integrates the explicit and implicit correlations of attribute labels. However, these hand-crafted relation rules could not be easy to create as the attribute tags vary in different datasets.

2.2 Class imbalance

The class imbalance problem has been investigated extensively in machine learning over the last two decades. Conventional imbalanced learning method, including sampling-based and cost-sensitive approaches [27], are proposed to balance class ratio or penalize classification errors of minority class [28]. Recently, deep imbalanced learning methods are also introduced to address this issue. CRL [29] provides a Class Rectification Loss (CRL) function to discover sparse minority classes boundaries for deep network architecture. CLMLE [30] enforces deep CNNs to learn the inter-cluster margins so that it can carve the more balanced class boundaries. However, the class imbalance has been underestimated in PAR since few works have been developed for this issue. DeepMar [5] applied a simple re-weighting strategy to adjust the loss weights between positive and negative classes during training. MsVAA [9] adopts a re-weighted loss to alleviate sample bias both at the class level and instance level with multi-scale visual attention.

2.3 Deep metric learning

Deep metric learning (also known as distance metric learning) [31] seeks for minimal difference between inter-class samples and maximal inter-class distances. Therefore, the classification ability of CNNs can be enhanced with metric learning functions, such as contrastive loss [32] and triplet loss [33], which are extensively employed in person Re-ID [34] and human face recognition [35]. HFE [36] uses the Person identity (ID) information in PAR to constraint the ID clusters for each attribute class to obtain a fine-grained feature embedding at the instance level. However, these supplementary personal IDs are not easily accessible in most PAR datasets.

3 Methodology

3.1 Problem definition

Given a labeled training PAR dataset $\mathcal{D}$ with N instances, denoted as $\mathcal{D} = \{x_i, \mathbf{y}_i\}$, where x_i is the ith sample and each instance has M labeled attributes, denoted as $\mathbf{y}_i = \{y_{i1}, y_{i2}, \cdots, y_{ij}, \cdots, y_{iM}\}$. y_{ij} is the individual label of jth attribute in x_i, and $y_{ij} \in \{0, 1\}$. The PAR dataset contains rich individual characteristics, thus we need to predict multiple classes y_{ij} simultaneously, where '1' means the corresponding attribute exists in the image and '0' denotes it is absent. To explore contextual relations among all the attributes, we jointly learn multiple binary classifiers with the shared backbone CNNs.

3.2 Neural network architecture

The overall framework of our model is illustrated in Fig. 2. The base architecture consists of two identical ResNet-50 backbones [37]. By randomly selecting two images as the inputs, we can obtain two deep feature maps from each identical CNN network that shares the same weights. Then, a global average pooling (GAP) layer is employed to concatenate these CNN features. Afterward, a batch normalization (BN) layer followed by the GAP is used to balance the outputs of positive and negative classes [24, 25, 38]. Then each branch of CNNs yields multiple prediction sores of M attributes in the range 0 to 1 with a sigmoid function. Finally, the training losses of both branches are calculated based on the proposed MCFL function, which consists of two basic modules, *i.e.*, multi-label focal loss and multi-label contrastive loss. The former emphasized the imbalanced minority classes. The later enhanced the discriminative features extraction by examining diverse attributes. No other alterations of the ResNet-50 are made except the points mentioned above.

3.3 Multi-label contrastive focal loss

3.3.1 Multi-label focal loss

Due to the uneven distribution of attributes in the PAR dataset, the classifier favors the negative majority class while ignoring those error-prone positive minority attributes. In this study, we want the classifier to pay more attention to the positive classes to learn more useful information. Here, we extend the Focal Loss [39] into the multi-label PAR tasks, and propose a multi-label Focal Loss (MFL), whose formula is defined below:

$$L_i^P = -\gamma_i (1 - \hat{\mathbf{y}}_i) \mathbf{y}_i \log(\hat{\mathbf{y}}_i) \tag{1}$$

$$L_i^N = \gamma_i \hat{\mathbf{y}}_i (1 - \mathbf{y}_i) \log(1 - \hat{\mathbf{y}}_i) \tag{2}$$

where L_i^P and L_i^N represent the positive-class loss and the negative-class loss for the ith instance, respectively. $\hat{\mathbf{y}}_i$ is multiple output scores of x_i. γ_i represents the sample weight at the instance level [5], which is applied to strengthen learning processes of minority attributes further and is defined as follows:

$$\gamma_i = \exp\left(\mathbf{y}_i \left(1 - \frac{\sum_{i=1}^N \mathbf{y}_i}{N} \right) + (1 - \mathbf{y}_i) \frac{\sum_{i=1}^N \mathbf{y}_i}{N} \right) \tag{3}$$

3.3.2 Multi-label contrastive loss

This section will introduce the proposed multi-label contrastive loss for PAR. Constructing proper distance metrics for multi-label tasks is challenging work. Unlike other pedestrian analysis tasks, *e.g.*, person re-ID [1] or face recognition [35], which can use the personal identity to restrict the same group of samples. In PAR, different persons have their own characters, which could be highly dissimilar from each other. Moreover, even the attributes of same person observed in different cameras could be various (*e.g.*, the man wearing glass or hat cannot be seen that clearly from the back view.), leading to significant changes in appearance. Therefore, constructing a uniform distance metric at the instance level is not easy in complex surveillance scenarios.

In this study, a novel multi-label contrastive loss (MCL) at the attribute level is proposed to address this issue. Our model trains on a input pair of images $\{x_i^a, x_i^b\}$ with multiple attribute tags $\mathbf{y}_i^a$ and $\mathbf{y}_i^b$, some examples can be listed as: $\mathbf{y}_i^a = [1, 0, 1, 1, ..., 1, 0, 1], \mathbf{y}_i^b = [0, 1, 0, 1, ..., 1, 0, 0]$. As shown in Fig. 3, we can define the distance metrics between the pairwise samples based on the divergence of

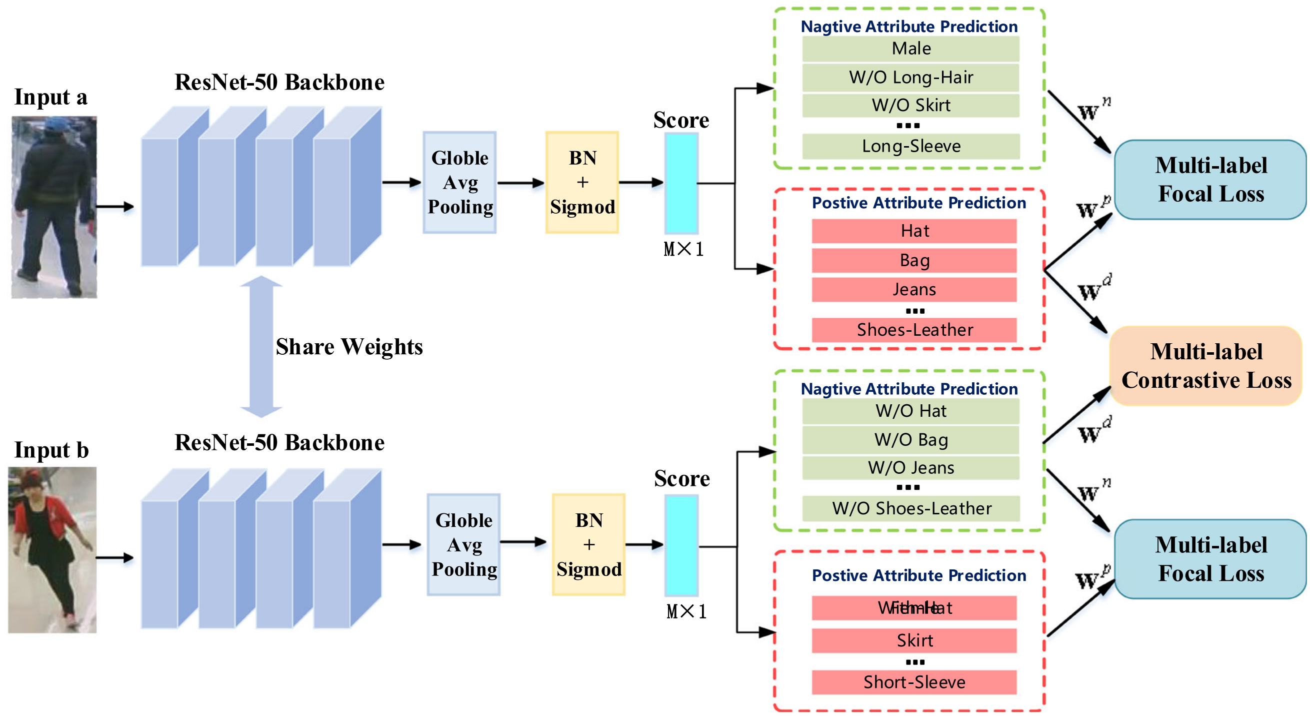


Fig. 2 The framwork of the proposed model

the attributes labels, so that we can enlarge gaps of inter-classes to enhance learning for more discriminant features.

First, a multi-label re-weighting mechanism is constructed to control the loss of each attribute. More specially, the attributes in training instances can be re-weighted according to the label diversity between the input pairs:

- $\mathbf{w}_i^p$ is the weight of the attributes sharing same positive tags.
- $\mathbf{w}_i^n$ is the weight of the attributes having same negative tags.
- $\mathbf{w}_i^d$ is the weight of the attributes with opposite tags.

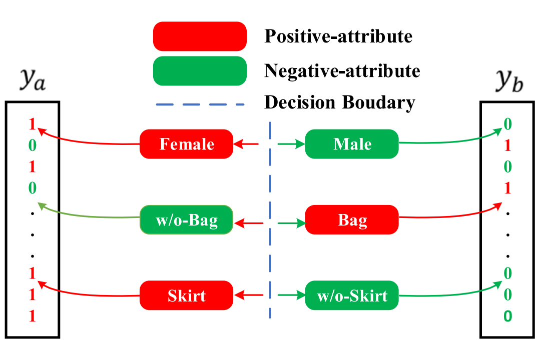


Fig. 3 An illustration of our multi-label contrastive learning. The decision boundary between dissimilar pairs with different labels is enlarged

$\mathbf{w}_i^p = \{w_{ij}^p\}_{j=1}^M$, $\mathbf{w}_i^n = \{w_{ij}^n\}_{j=1}^M$ and $\mathbf{w}_i^d = \{w_{ij}^d\}_{j=1}^M$ have the same shape as $\mathbf{y}_i^a$, $\mathbf{y}_i^b$. Here, $\mathbf{w}_i^p, \mathbf{w}_i^n, \mathbf{w}_i^d$ indicate which kind of attribute losses could be activated and backpropagated in the training process. The detailed computation is as follows.

$$w_{ij}^p = \begin{cases} 1, & y_{ij}^a + y_{ij}^b = 2 \\ 0, & others \end{cases} \tag{4}$$

$$w_{ij}^n = \begin{cases} 1, & y_{ij}^a + y_{ij}^b = 0 \\ 0, & others \end{cases} \tag{5}$$

$$w_{ij}^d = \begin{cases} 1, & y_{ij}^a + y_{ij}^b = 1 \\ 0, & othters \end{cases} \tag{6}$$

where $i \in [1, N]$ denotes the ith input pair, j denotes the jth attribute, and $j \in [1, M]$.

The cosine similarity is applied to measure the input pair-wise pedestrians using the following formulations:

$$d(\phi(x_a), \phi(x_b)) = \begin{cases} 1 - \dfrac{\sum_{j=1}^M \hat{y}_j^a \hat{y}_j^b w_j^d}{\sqrt{\sum_{j=1}^M (\hat{y}_j^a w_j^d)^2 \sum_{j=1}^M (\hat{y}_j^b w_j^d)^2}} & , |\mathbf{w}_j^d| > 0 \\ 0, & |\mathbf{w}_j^d| = 0 \end{cases} \tag{7}$$

where $\phi(\cdot)$ is the sigmoid output function, where the default classification threshold is set to 0.5, *i.e.*, if $\hat{y}_{ij} > 0.5$ means the prediction of the jth attribute appeared in ith image. The output of ϕ is the list of all the attributes, $\hat{y}^a_{ij}$ and $\hat{y}^b_{ij}$ are the output scores for jth attributes of ith paired samples $\{x^a_i, x^b_i\}$, respectively. $d(x^a_i, x^b_i)$ measures the cosine distance between the ith paired input images $\{x^a_i, x^b_i\}$ with multiple diverse attribute tags under the constrain of $\mathbf{w}^d_i$. Note that the greater the divergence between the input $\mathbf{y}^a_i$ and $\mathbf{y}^b_i$ is, *i.e.*, the input pairs have very little in common on visual attributes, the higher value of distance d could be generated. Otherwise, if the outputs of inter-class in $\mathbf{y}^a_i$ and $\mathbf{y}^b_i$ are similar to each other, the decision margin for the inter-class is closer.

The decision boundary should be more rigorous to achieve better classification results by yielding a more significant margin in the prediction scores between inter-class attributes. Hence, the ambiguous classes near the boundary line could be emphasized, and the backbone of CNNs can be encouraged to extract more discriminative deep features from the corresponding attribute-related regions. Meanwhile, the more difficult classification criteria can avoid over-fitting when learning with limited training instances.

The multi-label contrastive loss is given by

$$\mathcal{L}_{Cons\_P} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{w}^d_i L^P_i)[\alpha - d(\phi(x^a_i), \phi(x^b_i))] \tag{8}$$

$$\mathcal{L}_{Cons\_N} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{w}^d_i L^N_i)[\beta - d(\phi(x^a_i), \phi(x^b_i))] \tag{9}$$

Here, $\mathcal{L}_{Cons\_P}$ and $\mathcal{L}_{Cons\_N}$ are the inter-class constructive losses for positive and negative classes with different margins, respectively, while α and β adjust the margins of positive and negative attributes separately.

3.3.3 Total loss function

By combing the proposed MFL and MCL, our final MCFL function are listed follows:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{w}^p_i L^P_i + \mathbf{w}^n_i L^N_i) + L_{Cons\_P} + L_{Cons\_N} \tag{10}$$

MCFL consists of the following parts: the first two parts are the multi-label focal loss of the intra-class of attributes, while $\mathcal{L}_{Cons\_P}$ and $\mathcal{L}_{Cons\_N}$ are the multi-label constructive loss of the inter-class of attributes. We set the hyper-parameters α and β to 1.8 and 1, receptively. The effects of these two hyper-parameters will be discussed in the ablation experiment.

4 Experiments

4.1 Datasets

Richly Annotated Pedestrian (RAP) dataset [5]: includes 41,585 images captured with 26 cameras in the real indoor surveillance scenarios. RAP is split into 33,268 images for training and 8317 images for testing, where a total of 51 attributes are selected for evaluation [5].

PA100K [17] dataset: involves 100,000 pedestrian images collected from 598 cameras in the real outdoor surveillance scenarios. It is divided into training, validation, and testing with the ratio of 8:1:1, in which 26 commonly used binary attributes are tested for comparison.

PEdesTrian Attribute (PETA) dataset [40]: consists of ten small person re-ID datasets. It includes 8,705 people with 19,000 samples, which are divided into three partitions: 9500 images for training, 1900 images for validation, and 7600 images for testing. A total of 35 attributes are chosen considering whose proportion of positive labels are greater than 1/20 following [40].

RAP$_{zs}$ and PETA$_{zs}$ [38] datasets: are constructed under zero-shot pedestrian identity setting. The performance of PAR could be overestimated since the pedestrian identities are overlapped between the training set and the test set. To give reliable evaluations, RAP$_{zs}$ and PETA$_{zs}$ are constructed in [38] by removing the overlapped identities in RAP and PETA, respectively. The RAP$_{zs}$ contains only 26,638 images with selected 54 binary attributes, including 14,729 training images, 5,961 validation images, and 5,948 test images. PETA$_{zs}$ is partitioned into 11,051 for training, 3,980 for validation, and 3,969 for testing with the same 35 binary attributes according to [38].

Market-1501 and DukeMTMC attribute datasets [3]: are widely adopted benchmarks for person re-ID, thus their attributes information are manually added with the personal identity in [3]. The Market-1501 attribute dataset contains 32,668 samples of 1,501 identities, and in which each is labeled with nine binary attributes and three multi-class attributes (*i.e.*, age and two multi-class colors for upper and lower clothing). DukeMTMC attribute dataset contains 34,183 samples of 1,404 identities, and in which each includes eight binary attributes and two multi-class attributes (*i.e.*, two clothing colors).

4.2 Evaluation criteria

We usd two metrics, *i.e.*, label-based and the instance-based metrics, to validate the performance of PAR. The label-based meatric first assesses each attribute's accuracy separately, then the total mean accuracy (mA) is measured

Table 1 The comparison results on the RAP

Methods	Backbone	mA	Acc	Prec	Rec	F1
HPNet [17]	Inceptionnet	76.12	65.39	77.33	78.79	78.05
LGNet [16]	InceptionV2	78.68	68.00	80.36	79.82	80.09
PGDM [11]	Caffenet	74.31	64.57	78.86	75.90	77.35
JLPLS-PAA [19]	–	81.25	67.91	78.56	81.45	79.98
RA [24]	InceptionV3	81.16	–	79.45	79.23	79.34
AAP [41]	ResNet-50	81.42	68.37	**81.04**	80.27	80.65
ALM [42]	BNInception	81.87	68.17	74.71	86.48	80.16
DeepMar-R [5]	ResNet-50	77.38	66.58	79.87	77.98	78.91
IA2-Net [43]	ResNet-152	77.44	67.75	79.01	77.45	78.03
MTA [22]	ResNet-152	77.62	67.17	79.72	78.44	79.07
DA-HAR [8]	ResNet-101	79.44	**68.86**	80.14	81.30	**80.72**
HR-net [44]	ResNet-50	81.10	45.70	51.48	78.56	62.20
JLAC [25]	ResNet-50	81.51	66.29	77.17	80.42	78.76
SSC [45]	ResNet-50	82.77	68.37	75.05	87.49	80.43
MCFL(ours)	ResNet-50	**84.04**	67.28	73.44	**87.75**	79.96

Best results are highlighted in bold and the second-best results are underlined

Table 2 The comparison results on PA100K

Methods	Backbone	mA	Acc	Prec	Rec	F1
HPNet [17]	Inceptionnet	74.21	72.19	82.97	82.09	82.53
LGNet [16]	InceptionV2	76.96	75.55	86.99	83.17	85.04
PGDM [11]	Caffenet	74.95	73.08	84.36	82.24	83.29
AAP [41]	ResNet-50	80.56	**78.30**	**89.49**	84.36	**86.85**
ALM [42]	BNInception	80.68	77.08	84.21	**88.84**	86.46
Deepmar-R [5]	ResNet-50	79.03	76.08	85.60	85.27	85.43
MT-CAS [46]	ResNet-34	77.20	78.09	88.46	84.86	86.62
MCFL(ours)	ResNet-50	**81.53**	77.80	85.11	88.20	86.62

Best results are highlighted in bold and the second-best results are underlined

Table 3 The comparison results on PETA

Methods	Backbone	mA	Acc	Prec	Rec	F1
HPNet [17]	Inceptionnet	81.77	76.13	84.92	83.24	84.07
PGDM [11]	Caffenet	82.97	78.08	86.86	84.68	85.76
RA [24]	InceptionV3	86.11	–	84.69	88.51	86.56
MsVAA [9]	ResNet101	84.59	78.56	86.79	86.12	86.46
JLPLS-PAA [19]	–	84.88	79.46	87.42	86.33	86.87
IA2-Net [43]	ResNet-152	84.13	78.62	85.73	86.07	85.88
DeepMar-R [5]	ResNet-50	83.86	78.10	86.91	85.11	86.00
MTA [22]	ResNet-152	84.62	78.80	85.67	86.42	86.04
MT-CAS [46]	ResNet-34	83.17	78.78	**87.49**	85.35	86.41
SSC [45]	ResNet-50	86.52	78.95	86.02	87.12	86.99
CAS-SAL-FR [23]	ResNet-50	86.40	**79.93**	87.03	87.33	**87.18**
MCFL(ours)	ResNet-50	**86.83**	78.89	84.57	**88.84**	86.65

Best results are highlighted in bold and the second-best results are underlined

Table 4 The comparison results on RAP_{zs} and PETA_{zs}

Model	Backbone	RAP_{zs}					PETA_{zs}				
		MA	Acc	Pre	Recall	F1	MA	Acc	Pre	Recall	F1
MsVAA [9][1]	ResNet-50	$\underline{71.32}$	$\underline{63.59}$	77.22	76.62	76.44	71.03	59.38	$\underline{74.75}$	70.10	**72.37**
VAC [47][1]	ResNet-50	70.20	**65.45**	**79.87**	76.65	**77.07**	71.05	58.90	**74.98**	70.48	$\underline{72.13}$
ALM [42]	BN-Inception	70.62	62.78	74.52	$\underline{77.97}$	76.20	71.41	58.13	69.94	$\underline{74.21}$	72.02
DeepMar-R [5]	ResNet-50	68.01	62.21	$\underline{77.60}$	73.84	75.67	70.25	58.05	74.14	69.84	71.92
MCFL(ours)	ResNet-50	**74.37**	63.37	71.21	**83.86**	$\underline{77.02}$	**72.91**	57.04	68.47	**74.35**	71.29

Best results are highlighted in bold and the second-best results are underlined

[1]Is re-implemented by [38] using the ResNet-50 for a fair comparison

Table 5 The comparison results on Market-1501. "L.low", "L.slv", "U.color" and "L.color" mean length of lower-body clothing, sleeve length, color of upper-body clothing and color of lower-body clothing, respectively

Methods	Age	Backpack	Bag	handbag	Clothes	L.low	L.slv	Hair	Hat	gender	U.color	L.color	Avg.
DeepMar [5]	**91.5**	83.8	60.8	$\underline{89.3}$	89.1	88.6	93.5	82.8	97.1	82.6	71.1	66.9	83.1
APN [3]	85.8	86.6	**78.6**	88.1	93.6	93.6	93.5	84.2	97.0	87.5	72.4	71.7	86.1
APR [3]	88.6	$\underline{84.9}$	76.4	**90.4**	92.8	$\underline{93.7}$	**93.6**	84.4	97.1	88.9	$\underline{74.0}$	**73.8**	$\underline{86.6}$
VAC [47]	88.0	**87.5**	$\underline{76.7}$	88.1	$\underline{94.2}$	93.3	93.5	$\underline{88.3}$	$\underline{97.3}$	$\underline{91.4}$	69.5	64.6	86.0
MCFL	$\underline{90.1}$	84.4	75.4	85.0	**94.3**	**95.1**	93.5	**88.8**	**97.4**	**91.5**	**76.9**	$\underline{73.6}$	**87.2**

Best results are highlighted in bold and the second-best results are underlined

Table 6 The comparison results on DukeMTMC. "L.slv", "U.color" and "L.color" mean sleeve length, color of upper-body clothing and color of lower-body clothing, respectively

Methods	Backpack	Bag	Handbag	Boots	Gender	Hat	Shoes	L.slv	U.color	L.color	Avg.
DeepMar [5]	77.1	$\underline{82.3}$	93.6	86.5	84.6	87.0	$\underline{90.0}$	85.5	67.9	62.1	81.7
APN [3]	$\underline{77.5}$	82.2	92.3	88.3	82.0	85.5	87.6	86.2	73.4	68.3	82.3
APR [3]	75.8	**82.9**	$\underline{93.4}$	87.5	84.2	87.6	89.7	**88.4**	$\underline{74.2}$	**69.9**	83.4
VAC [47]	**82.2**	81.6	**93.6**	$\underline{90.0}$	**85.8**	$\underline{88.1}$	**90.8**	$\underline{88.1}$	71.8	68.3	$\underline{84.0}$
MCFL	**82.2**	81.0	93.3	**90.3**	$\underline{85.1}$	**89.0**	$\underline{90.0}$	86.8	**76.0**	$\underline{69.3}$	**84.3**

Best results are highlighted in bold and the second-best results are underlined

over all the attributes as the evaluation indicator, which is shown as follows:

$$mA = \frac{1}{2N}\sum_{i=1}^{M}\left(\frac{TP_i}{P_i} + \frac{TN_i}{N_i}\right) \tag{11}$$

where P_i and TP_i are the counts of real positive attributes and properly predicted positive attributes of the ith instance, respectively, in contrast, N_i and TN_i the number of real negative attributes and the number of correctly predicted negative attributes of the ith instance, respectively. N is the total number of examples.

The instance-based metrics use four conventional criteria to assess the prediction results on each image, including accuracy, recall, precision, and F1-measure, which are given as:

$$Acc_{example} = \frac{1}{N}\sum_{i=1}^{N}\frac{|Y_i \bigcap f(x_i)|}{|Y_i \bigcup f(x_i)|} \tag{12}$$

$$Prec_{example} = \frac{1}{N}\sum_{i=1}^{N}\frac{|Y_i \bigcap f(x_i)|}{|f(x_i)|} \tag{13}$$

$$Rec_{example} = \frac{1}{N}\sum_{i=1}^{N}\frac{|Y_i \bigcap f(x_i)|}{|Y_i|} \tag{14}$$

$$F1 = \frac{2 * Prec_{example} * Rec_{example}}{Prec_{example} + Rec_{example}} \tag{15}$$

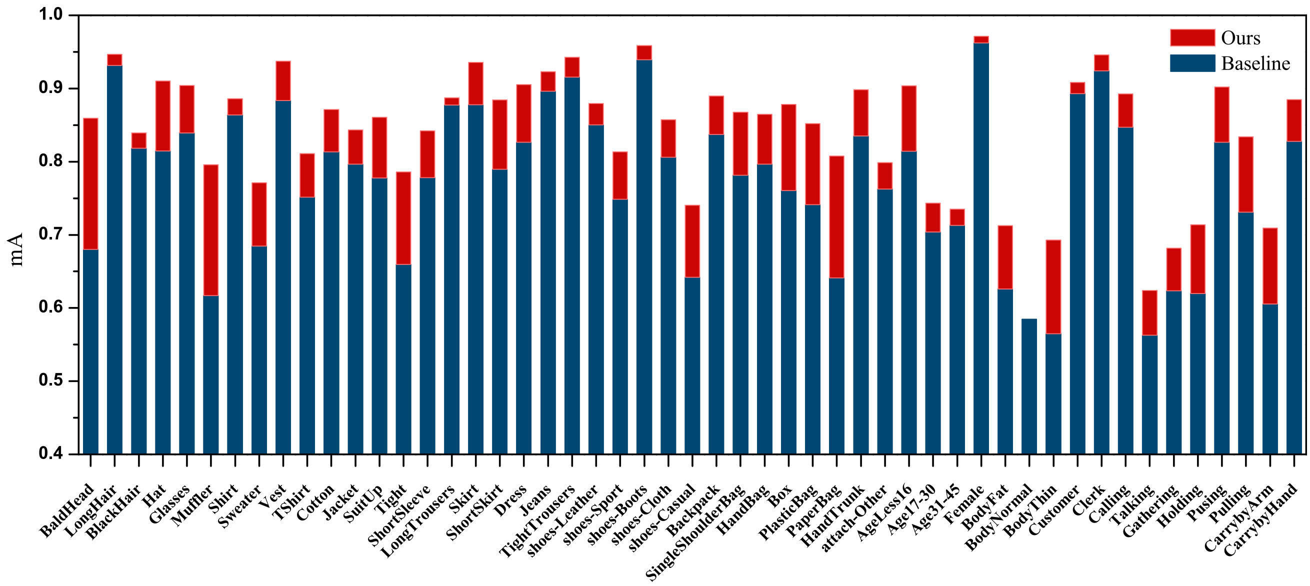


Fig. 4 The mA results of each attribute with MCFL and baseline model on RAP dataset

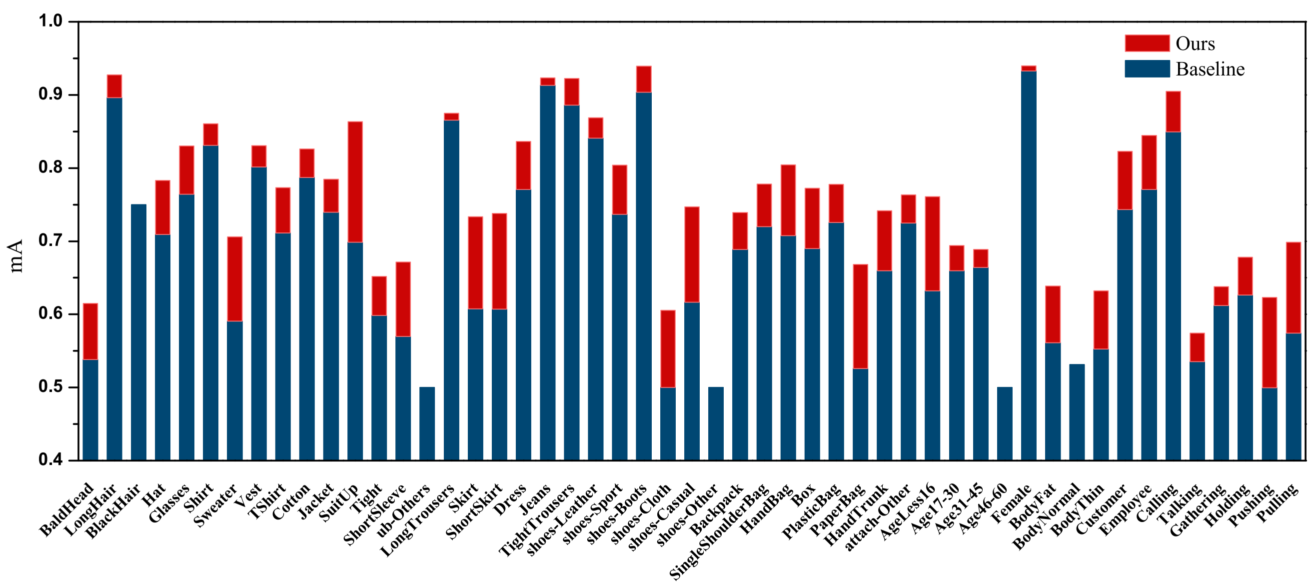


Fig. 5 The mA results of each attribute with MCFL and baseline model on RAP_{zs} dataset

where Y_i denotes the ture positive attribute labels, and $f(x_i)$ is the actually predicted positive labels. F1-measure is the trade-off between the precision and recall vaules.

4.3 Training details

In our model, the ResNet-50 is adopted as the backbone CNNs in our multi-branch architectures with the input image size of 256 x 192. Several image augmentation operations are used to enhance the model training, such as random erasing, random cropping, and horizontal flipping. The Stochastic gradient descent (SGD) is employed as the optimizer of MCFL with a learning rate of 0.001 and weight decay of 0.005 and mini-batch size of 64. The proposed model is trained for 40 epochs on dual NVIDIA GTX1080Ti with the Pytorch framework.

4.4 Comparison to state-of-the-art

The proposed MCFL are compared with several state-of-the-art (SOTA) approaches applied in PAR, including body-part-based methods, *e.g.*, PGDM [11], ALM [42], and AAP [41]; visual attention-based models, *e.g.*, HPNet [17], VAC [47], MT-CAS [46], JLPLS-PAA [19], CAS-

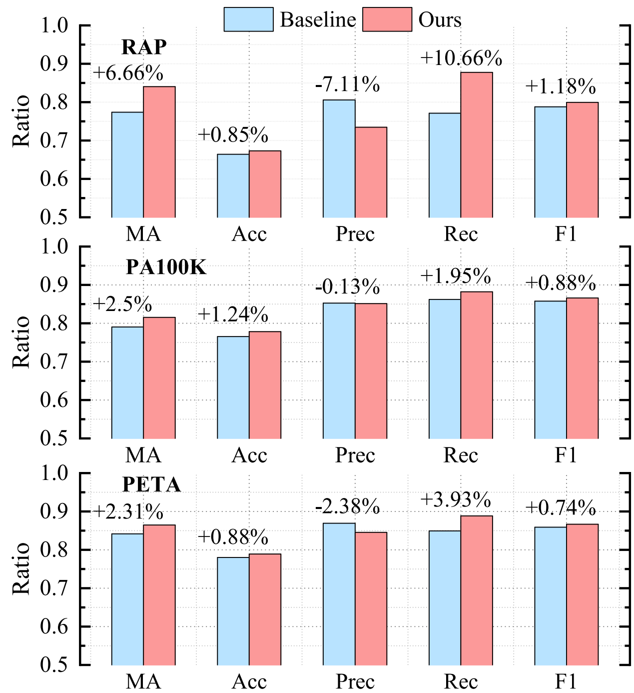


Fig. 6 The experiment results of our method comparing with Baseline on PA100K, RAP and PETA. We visualize the mA, ACC, Prec, Rec and F1 values, receptively

SAL-FR [23], and DA-HAR [8]; Relation-Based models, *e.g.*, RA [24], IA2-Net [43], MTA-Net [22] JLAC [25], SSC [45] and HR-net [44]; other loss function based models, *e.g.*, DeepMar-R [5], MsVAA [9], APN [3] and APR [3].

It needs to be pointed out that the same backbone of ResNet-50 and sample weights strategy are adopted in our model as DeepMar-R [5], thus it is used as the baseline model in the experiments. The PAR results are presented in Tables 1, 2 and 3 for RAP, PA100K and PETA datasets, respectively, and Table 4 for RAP_{zs} and $PETA_{zs}$, along with Tables 5 and 6 for Market-1501 and DukeMTMC, respectively. Note that, we highlighted the best and second-best results in bold and underlined in these tables.

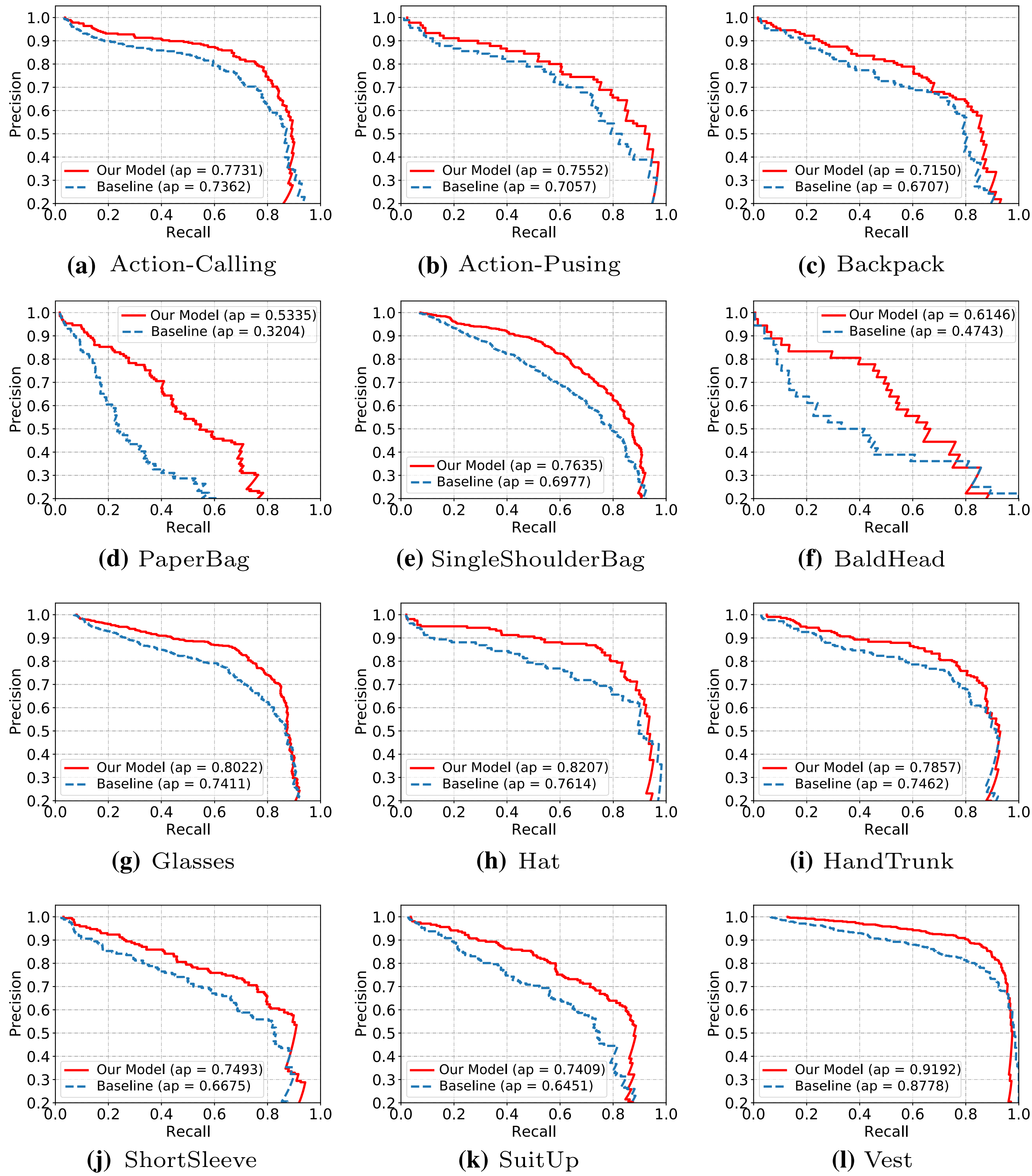
As we can see from Tables 1, 2 and 3, our MCFL achieves the best performances on RAP, PETA datasets in both mA and recall metrics with comparable F1 values. It also generates the best mA values, second-best recall, and F1 values on the PA100K dataset. More specifically, on the RAP dataset, MCFL outperforms SSC [45], which achieves the second-best mA result, MCFL achieves an mA gain of 1.27%, along with a 0.26% recall raise. It could be regarded as a promising improvement as most of the tested SOTA methods obtain mA values less than 82%, owing to the main challenges such as low-quality images, highly-skewed attributes distribution, and so on [48]. Besides, most SOTA methods that rely on additional visual attention and parts-location modules, or other external hand-crafted relationship rules, employ even much deeper CNNs backbones (*e.g.*, ResNet-152 and ResNet-101 are used in IA2-Net [43] and DA-HAR [8], respectively). However, our model using the naive ResNet-50 backbone with fewer parameters can still achieve promising results. These further demonstrate that our proposed MCFL provided a possible way to improve the PAR with the optimized loss function.

Turning to the more challenging RAP_{zs} and $PETA_{zs}$ with the zero-shot setting in Table 4, we can infer that the performance of all the tested approaches dropped significantly. However, our proposed model still obtains the highest mA and produces much higher recall values, thus achieves the second-best F1 values on RAP_{zs}. The improvement on $PETA_{zs}$ is smaller as the problem of class imbalance in $PETA_{zs}$ is less severe. Our method still reaches better results in terms of mA compared to existing methods.

In addition, as we can seen from Tables 5 and 6, our method consistently produces the best values of total average accuracy among all the tested methods on Market-1501 and DukeMTMC, *e.g.*, compared with APR [3] that combines PAR with person identity classification, our method results in 0.6% and 0.9% improvements of total average accuracy on Market-1501 and DukeMTMC, respectively.

Comparing to the baseline model (DeepMar-R [5]), which also adapts the weighted BCE-loss assigning penalties to emphasize the minority attributes. As shown in Figs. 4 and 5, our MCFL produces better mA values than the baseline model on most attributes, especially on some fine-grained attributes, such as shoes, bags, and muffler. As can be observed from Fig. 6, our method also generates better prediction results than the baseline model, improving by 6.66% and 9.77% on RAP, 2.5% and 2.93% on PA100K, 2.97% and 3.73% on PETA, in mA and recall, respectively. All of these demonstrate that our approach could be more suitable for pedestrian attributes retrieving in practice.

To further assess the performance of our MCFL on imbalanced datasets, we present the Precision-Recall (PR) curves of the attributes with the uneven distributions, which are shown in Fig. 7a-(l), along with the average precision (ap) values that summarize the precision-recall trade-off [49]. It can be informed from Fig. 7 that our model consistently obtains the higher area under the PR curve than that of the baseline model, which represents that our model can generate both higher recall and precision values, especially performs better on some tail categories, *e.g.*, BaldHead, PaperBag, Glass, and Hat. Furthermore, those minority positive classes of attributes could be important clues for searching the particular persons.



(a) Action-Calling **(b)** Action-Pusing **(c)** Backpack

(d) PaperBag **(e)** SingleShoulderBag **(f)** BaldHead

(g) Glasses **(h)** Hat **(i)** HandTrunk

(j) ShortSleeve **(k)** SuitUp **(l)** Vest

Fig. 7 The Precision-Recall curves of the attributes with an uneven distribution on the RAP dataset. “ap” means the average precision value that summarizes the area under the precision-recall curve

Examples of attributes prediction results based on the baseline and proposed models are given in Fig. 8. We observe that both models can successfully recognize most of the attributes, even pedestrians in the image are incomplete. Our method delivers better performance than baseline, which neglected some fine-grained classes,

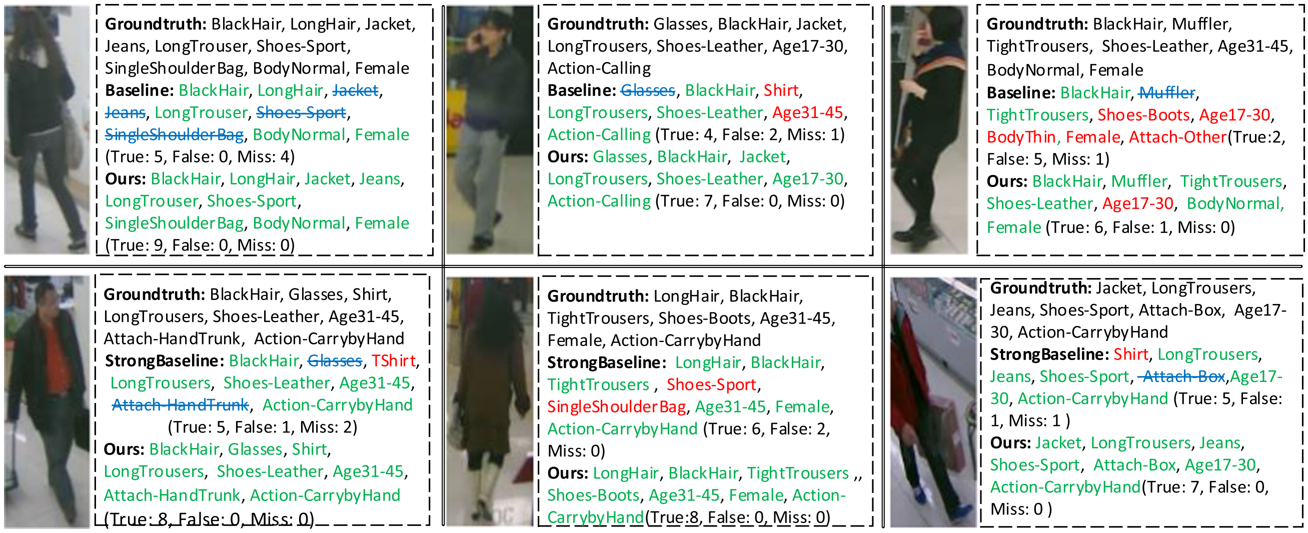


Fig. 8 Some examples of attributes recognition on RAP with our MCFL and baseline. Attributes in green represent the correct predictions, while the red labels mean error predictions compared to the ground truth list, and the blues in strikeout mean the prediction attributes are missed by the tested model compare with the groundtruth list. Those recognition results are counted using Ture, False, and Miss, respectively

Table 7 The resuls of different losses based on the same ResNet-50 backonbe on the RAP dataset

Methods	mA	Acc	Prec	Rec	F1
Baseline	77.38	66.58	79.87	77.98	78.91
MFL	80.11	67.92	79.43	80.42	79.55
MFL+$\mathcal{L}_{Cons\_P}$	83.59	63.79	66.74	**92.93**	77.68
MFL+$\mathcal{L}_{Cons\_N}$	81.56	**69.57**	80.09	82.34	**81.20**
MCFL(ours)	**84.04**	67.28	73.44	87.75	79.96

Best results are highlighted in bold

including bag, hat, carrying-boxes, and shoe-types. However, these characteristics could be helpful in practice. Meanwhile, the age range is more difficult to predict accurately as its confusing appearance.

4.5 Ablation study

4.5.1 Effects of components

We conduct further ablation studies to showcase the effectiveness of MCFL, *i.e,*, we train the PAR models with the same ResNet-50 backbone using the different losses. The results are given in Table 7. (1) WBCE is the weighted binary cross-entropy loss in baseline [5]. (2) MFL is the multi-label focal loss. (3) $\mathcal{L}_{Cons\_P}$ is the positive-class constructive loss. (4) $\mathcal{L}_{Cons\_N}$ is the negative-class constructive loss. (5) MCFL is the proposed method. These models are implemented with the same experimental settings.

Compared with the traditional WBCE loss function, MFL improves the mA and recall values by 2.89% and 2.29%, receptively. $\mathcal{L}_{Cons\_P}$ further achieve a better performance, performance, improving 3.48% and 12.51% in

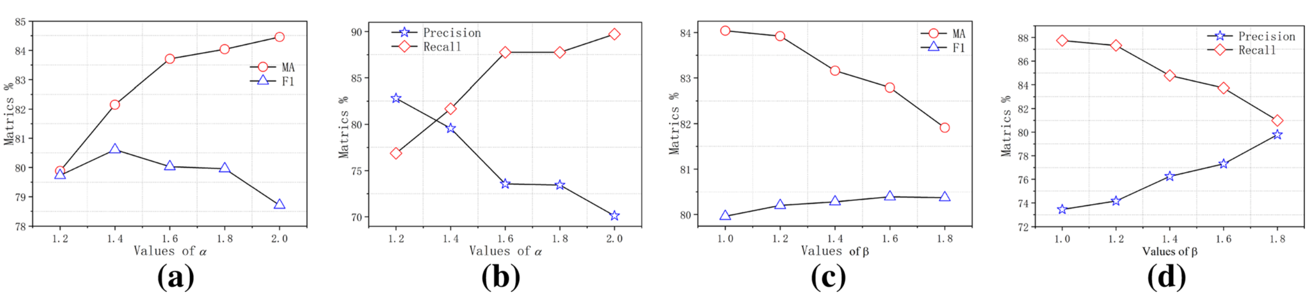


Fig. 9 Visualization of the impact of α and β. **a**The impact of α in terms of mA and F1. **b** The impact of α in terms of precision and recall. **c**The impact of β in terms of mA and F1. **d** The impact of β in terms of precision and recall

Table 8 The effcts of backbones on the RAP dataset

Model	MA	Acc	Prec	Recall	F1
ResNet-34	82.13	66.72	71.94	89.04	79.58
ResNet-50	84.04	**67.28**	**73.44**	87.75	**79.96**
ResNet-101	84.16	66.68	71.94	89.04	79.58
ResNet-152	84.76	66.92	72.23	89.06	79.76
DenseNet-161	84.74	66.95	72.51	88.63	79.76
DenseNet-169	**84.82**	66.02	70.93	**89.49**	79.13

Best results are highlighted in bold

mA and recall metrics over MFL. However, the losses of the majority attribute classes are suppressed, leading to a significant drop in precision. $\mathcal{L}_{Cons\_N}$ loss, which emphasizes the majority attributes, delivers the highest accuracy of 69.57% and F1 of 81.20%. In the end, the MCFL balances the precision and recall obtains the highest mA score of 84.04%.

4.5.2 Effects of hyper-parameters

We conduct the ablation experiment to assess the effects of hyper-parameters on RAP datasets. The hyper-parameters used in our model include α and β are used to adjust the margins between positive and negative classes. The results are demonstrated in Fig. 9. We first set $\beta = 1$ and vary the values of α and then set $\alpha = 1.8$ and vary the values of β. As we can inform from that, the larger α produces the higher values of mA and recall. However, the performances in terms of F1 and precision are significantly dropped down as $\alpha > 1.8$. While the performances of the model in terms of mA and recall are significantly reduced using a greater β, along with a slight improvement on F1 values. Finally, the model performs best when α and β is set to 1.8 and 1.

4.5.3 Effects of backbones

We evaluate our method using different the backbone networks on the RAP dataset, including ResNets [50] and DensNets [51]. The effects of backbones are shown in Table 8. As we can inform from that, the deeper backbones generally deliver the higher mA values due to their better capacity of feature representation. However, the improvement is limited, especially considering the computation cost and risk of over-fitting with large CNNs. Moreover, our method with ResNet-50 can perform best in both accuracy and precision, along with the highest F1 values. Thus, we adopt the ResNet-50 backbone in our PAR experiment.

5 Conclusion

This paper proposes a novel optimization algorithm for improving PAR. The proposed multi-label contrastive focal loss emphasizes the imbalanced minority classes and enhances the recognition capability of fine-grained attributes by exploring more discriminative features. The comprehensive experiments are conducted on five large PAR datasets. Our proposed MCFL with the naive ResNet-50 backbone can achieve better mA and recall values than the tested SOTA models, making it more suitable in practical scenarios. The designed efficient multi-label contrastive learning strategy can benefit many other multi-label learning tasks with uneven data distribution.

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Declarations

Conflict of interest All authors declared no conflict of interests.

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